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MAPPING TO SDG GOALS

The SmartPCB-XAI project aligns with the following United Nations Sustainable Development Goals (SDGs):

1. **SDG 9 – Industry, Innovation and Infrastructure:** Uses deep learning and explainable AI for automated PCB defect detection, supporting industrial innovation, improved quality inspection, and reliable electronics manufacturing.
2. **SDG 12 – Responsible Consumption and Production:** Supports early identification of PCB defects, helping reduce material waste, avoid unnecessary rework, and promote more efficient use of manufacturing resources.
3. **SDG 8 – Decent Work and Economic Growth:** Automates parts of the PCB inspection process, potentially reducing repetitive manual inspection effort and supporting productivity in electronics manufacturing.
4. **SDG 4 – Quality Education:** Provides a practical application of deep learning, computer vision, explainable AI, and digital twin technologies, supporting technical learning and skill development.